

Team Name : Astrothinkers

Team Leader Name : Atharv Kulkarni

Problem Statement : **Fill in the Frames Seamlessly - Enhancing Temporal Resolution of Satellite Imagery using the techniques of AI/ML based on Optical Flow**

Team Members

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Opportunity – How Is It Different?

How is it different from existing ideas?

Existing Optical Flow based interpolation methods regenerate the complete intermediate frame even when a large portion of the prediction is already reliable — often causing blur, ghosting and loss of temporal consistency during rapidly changing cloud movements.

Our approach introduces a confidence-guided reconstruction strategy. The model first estimates which regions are reliable and which are uncertain. Only the uncertain regions are refined using temporal information from neighbouring frames and meteorological parameters, while reliable regions are preserved.

This allows the model to generate more realistic intermediate satellite images without unnecessarily modifying correct information.

Opportunity – Solution Approach

How will it solve the problem?

Two consecutive satellite images are taken as input, and the motion between them is estimated using Optical Flow. Based on this motion, an initial intermediate frame is generated.

The generated frame is passed through a confidence estimation module that identifies regions where the prediction is uncertain. Only these regions are reconstructed using temporal context together with weather parameters such as cloud movement, wind speed and humidity.

Finally, the refined image is validated for temporal consistency before being used as the final intermediate satellite frame.

Opportunity – USP

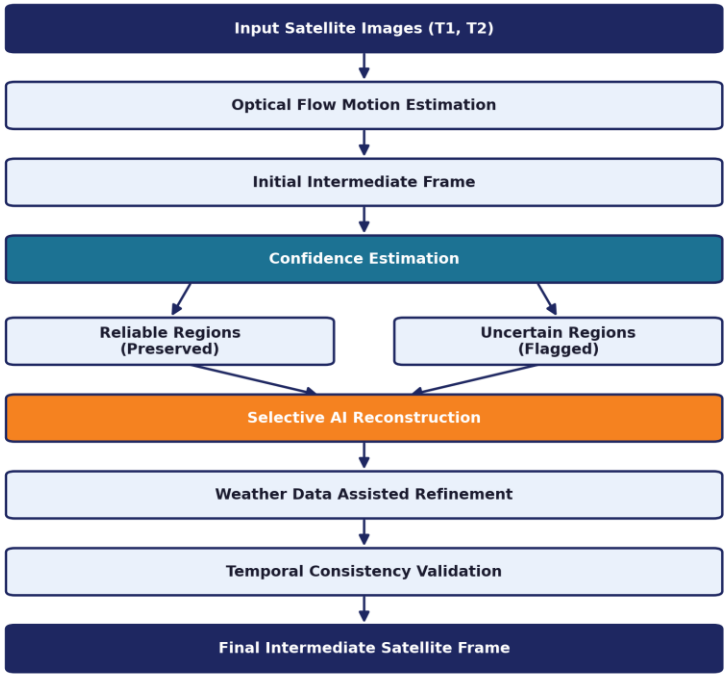
USP of the Proposed Solution

- Confidence-guided selective reconstruction instead of complete frame regeneration
- Preserves reliable observations while improving only uncertain regions
- Uses weather information only where additional refinement is required
- Reduces blur and ghosting during rapidly evolving cloud dynamics
- Produces more reliable intermediate satellite imagery for weather monitoring applications

List of Features Offered

- Optical Flow based motion estimation
- AI-based intermediate frame generation
- Pixel-level confidence estimation
- Selective reconstruction of uncertain regions
- Weather-assisted refinement
- Temporal consistency validation
- Confidence heatmap generation
- Image quality evaluation using standard metrics
- Interactive visualization of generated intermediate frames

Process Flow Diagram

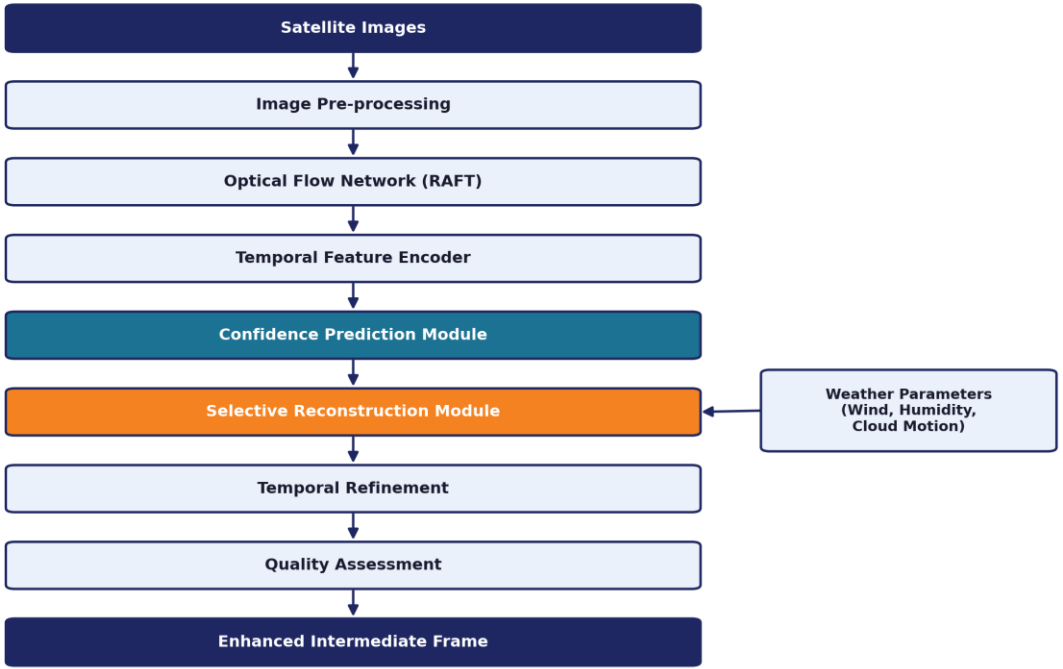


Wireframes / Mock Diagrams (Optional)

Wireframes are not included for this submission, as the proposed solution is primarily a backend AI/ML processing pipeline rather than a user-facing application.

A lightweight Streamlit/React dashboard (see Technologies Used) will be used to visualize input frames, confidence heatmaps and the generated intermediate satellite imagery.

Architecture Diagram



Technologies Used

Programming Language:

Python

AI / Deep Learning:

PyTorch, OpenCV, RAFT Optical Flow, Vision Transformer

Remote Sensing:

Rasterio, GDAL

Weather Data:

ERA5 Reanalysis Dataset, INSAT / Himawari Satellite Imagery

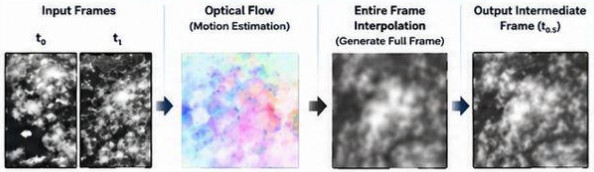
Visualization:

Streamlit, React

EXISTING TECHNOLOGY vs OUR PROPOSED SOLUTION

Goal: Generate high-quality intermediate satellite frames between two consecutive images

EXISTING OPTICAL FLOW INTERPOLATION



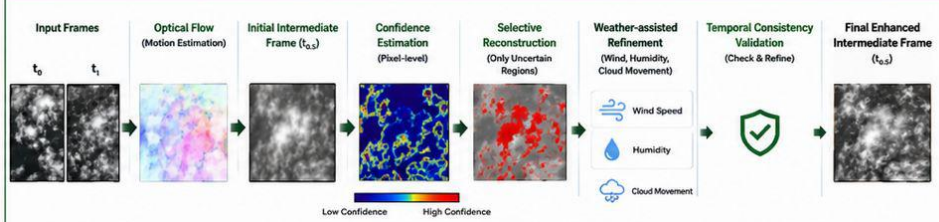
How Existing Methods Work

- Optical Flow is used to estimate motion.
- The entire intermediate frame is regenerated.
- Every pixel is treated equally.
- Often leads to blur, ghosting and temporal inconsistency, especially in fast-changing cloud regions.

Limitations

- ✖ Regenerates whole frame including reliable regions
- ✖ No distinction between reliable and unreliable pixels
- ✖ Does not use weather information
- ✖ Blur and ghosting in rapid cloud movement
- ✖ Higher computation on entire image

OUR PROPOSED SOLUTION (CONFIDENCE-GUIDED RECONSTRUCTION)



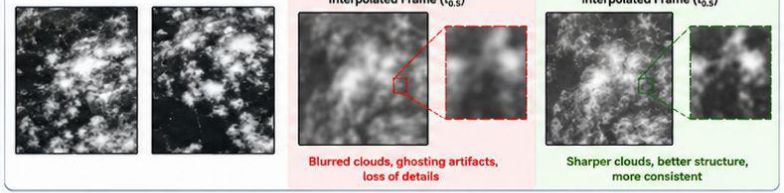
How Our Solution Works

- ✓ Estimates confidence for every pixel.
- ✓ Preserves reliable regions as they are.
- ✓ Reconstructs only uncertain regions using temporal context and weather parameters.
- ✓ Reduces blur and ghosting in fast-changing clouds.
- ✓ Improves temporal consistency and overall realism.

Key Advantages

- Confidence-guided selective reconstruction
- Preserves reliable observations
- Uses weather data only where needed
- Better temporal consistency
- Lower unnecessary computation
- Sharper, more realistic intermediate frames

VISUAL COMPARISON (EXAMPLE)



KEY DIFFERENCE AT A GLANCE

Aspect	Existing Technology	Our Proposed Solution
Frame Generation	Regenerates entire intermediate frame	Reconstructs only uncertain regions
Pixel Treatment	All pixels treated equally	Confidence score for every pixel
Use of Additional Data	Uses only image motion	Uses weather parameters for refinement
Artifacts	More blur and ghosting	Reduced blur, better consistency
Computation	High (whole frame processed)	Lower (only uncertain regions processed)
Output Quality	Lower in fast-changing regions	Higher quality and more realistic

Key Innovation: Confidence-guided selective reconstruction + Weather-assisted refinement ➤ **Better Temporal Resolution for Reliable Weather Monitoring**

Estimated Implementation Cost

The proposed solution primarily uses open-source software frameworks and publicly available satellite datasets.

Development can be carried out on standard GPU-enabled systems or educational cloud platforms.

Therefore, the implementation cost is expected to be minimal during the prototype stage.


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HACKATHON
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THANK YOU